

CONFOUNDING: OVERLOOKED IN BIOMEDICAL ANALYSES

Garrett Bullock, PT, DPT, DPhil

What is Confounding

Confounding: The relationship between an independent variable (exposure) and a dependent variable (outcome) is distorted by the presence of another variable.

1. Is associated with the exposure.
2. Is independently associated with the outcome.
3. Is not an intermediate step in the causal pathway between exposure and outcome.

Confounding Distorts Clinical Observations

► [Int J Epidemiol.](#) 2015 Jun 16;45(3):929–939. doi: [10.1093/ije/dyv104](#) [↗](#)

Coffee consumption and incidence of lung cancer in the NIH-AARP Diet and Health Study

[Kristin A Guertin](#)^{1,*}, [Neal D Freedman](#)¹, [Erikka Loftfield](#)¹, [Barry I Graubard](#)², [Neil E Caporaso](#)³, [Rashmi Sinha](#)¹

► [Author information](#) ► [Article notes](#) ► [Copyright and License information](#)

PMCID: PMC5005936 PMID: [26082405](#)

Confounding Distorts Clinical Observations

- Without accounting for confounding, cannot isolate the true effect of a relationship
- Directly effects clinical inferences

Three Stages of Identifying Confounders

Breast Cancer Research and Treatment **52**: 289-303, 1998.

© 1998 Kluwer Academic Publishers. Printed in the Netherlands.

Methodological challenges in the evaluation of prognostic factors in breast cancer

Douglas G. Altman¹ and Gary H. Lyman^{2,3}

¹*Imperial Cancer Research Fund Medical Statistics Group, Centre for Statistics in Medicine, Institute of Health Sciences, Oxford, UK;* ²*Medical Statistics Unit, Department of Epidemiology and Population Health, London School of Hygiene and Tropical Medicine, London, UK;* ³*H Lee Moffitt Cancer Center and Research Institute at the University of South Florida, Tampa, Florida, USA*

Three Stages of Identifying Confounders

1. Begins with expert knowledge

2. Addition of literature

3. Potential mechanisms

Need to Identify Which Type of Analyses You are Performing

1. Association: Describe
2. Predict: Forecast
3. Causal : Cause-effect

FULL ACCESS • VIEWPOINT

Description, Prediction, and Causation in Sport and Exercise Medicine Research: Resolving the Confusion to Improve Research Quality and Patient Outcomes

AUTHORS ^

Justin M. Losciale, PT, DPT^{1,2}, Garrett S. Bullock, PT, DPT, DPhil^{3,4,5}, Gary S. Collins, PhD^{6,7},

Amelia J.H. Arundale, PT, DPT, PhD^{8,9}, Tom Hughes, PT, PhD^{10,11}, Nigel K. Arden, MD³, Jackie L. Whittaker, PT, PhD^{1,2}

AFFILIATIONS v

Three Stages of Identifying Confounders

1. Univariable Analyses
2. Add Potential Covariates
3. Externally Replicate and/or Evaluate in Different Scenarios

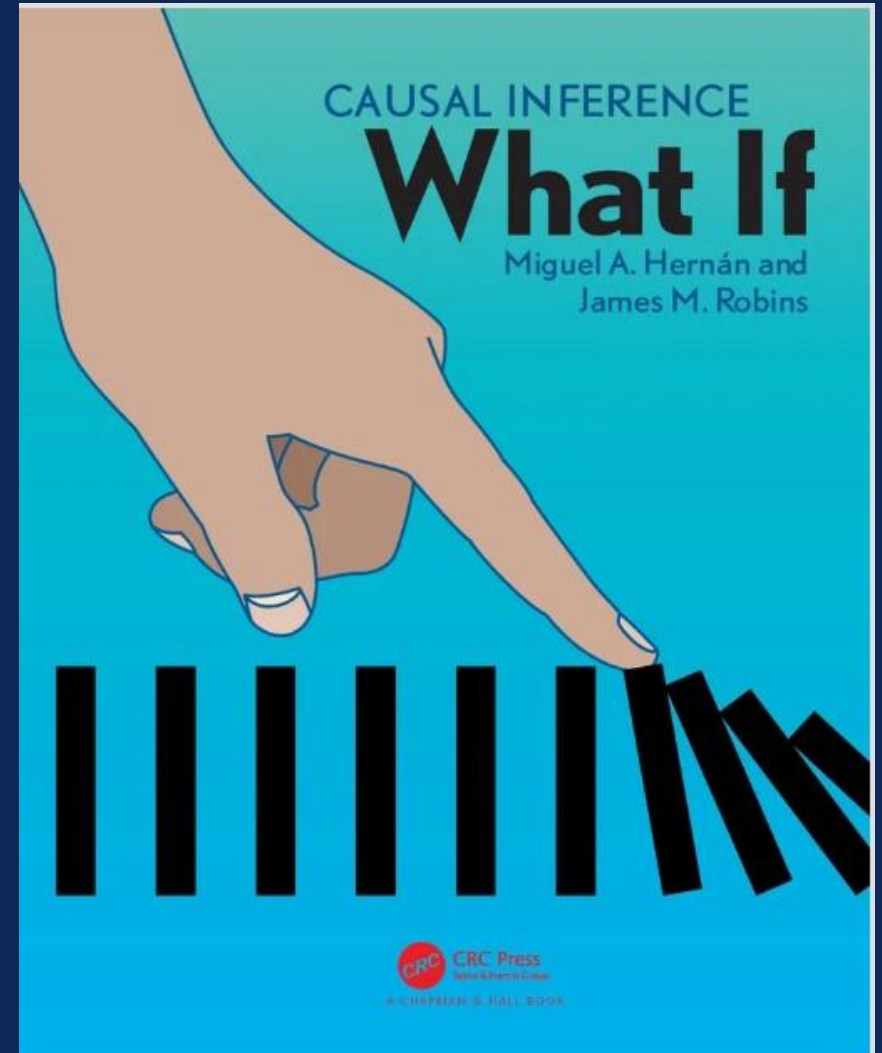
[VIEWPOINT]

JUSTIN M. LOSCIALE, PT, DPT^{1,2} • GARRETT S. BULLOCK, PT, DPT, DPhil^{3,4,5} • GARY S. COLLINS, PhD^{6,7}
AMELIA J.H. ARUNDALE, PT, DPT, PhD^{8,9} • TOM HUGHES, PT, PhD^{10,11} • NIGEL K. ARDEN, MD³ • JACKIE L. WHITTAKER, PT, PhD^{1,2}

Description, Prediction, and Causation in
Sport and Exercise Medicine Research:
Resolving the Confusion to Improve
Research Quality and Patient Outcomes

Confounding Should Be Theory & Empirical Driven NOT Data Driven

- Individual data – can be biased
- Random error
- Need to focus on overarching scientific literature



Confounding Should Be Theory & Empirical Driven NOT Data Driven

- No 'Causal Salad' or 'Factor's Associated with
- No forward or backward stepwise regressions
- Forget 'Table 2 Fallacy'

► [BMJ Med.](#) 2025 Oct 27;4(1):e001375. doi: [10.1136/bmjmed-2025-001375](https://doi.org/10.1136/bmjmed-2025-001375) [↗](#)

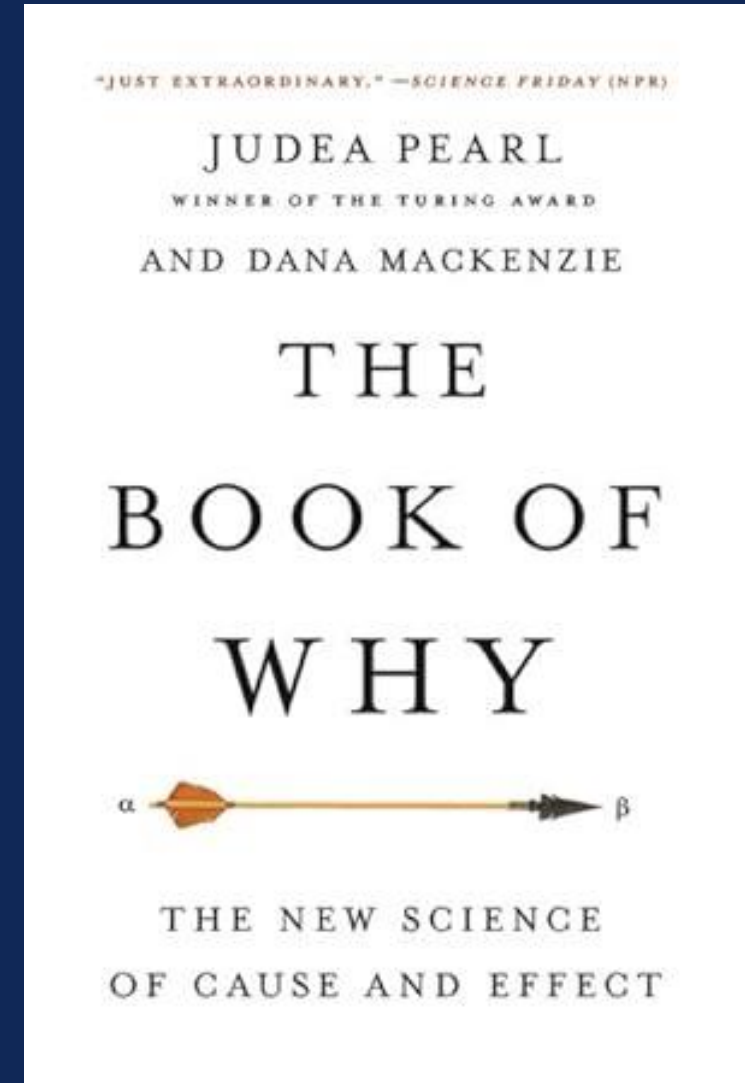
Factors associated with: problems of using exploratory multivariable regression to identify causal risk factors

[Dan Lewer](#)^{1,2,✉,0}, [Thomas Brothers](#)^{3,0}, [Elizabeth O'Nions](#)^{1,0}, [John Pickavance](#)^{1,0}

▼ [Author information](#) ► [Article notes](#) ► [Copyright and License information](#)

How Do We Create These Confounding Structural Relationships?

- Search & synthesize the literature
- Evaluate biological plausibility
- Interview experts (if necessary)
- Create a structural diagram to explicitly detail this relationship

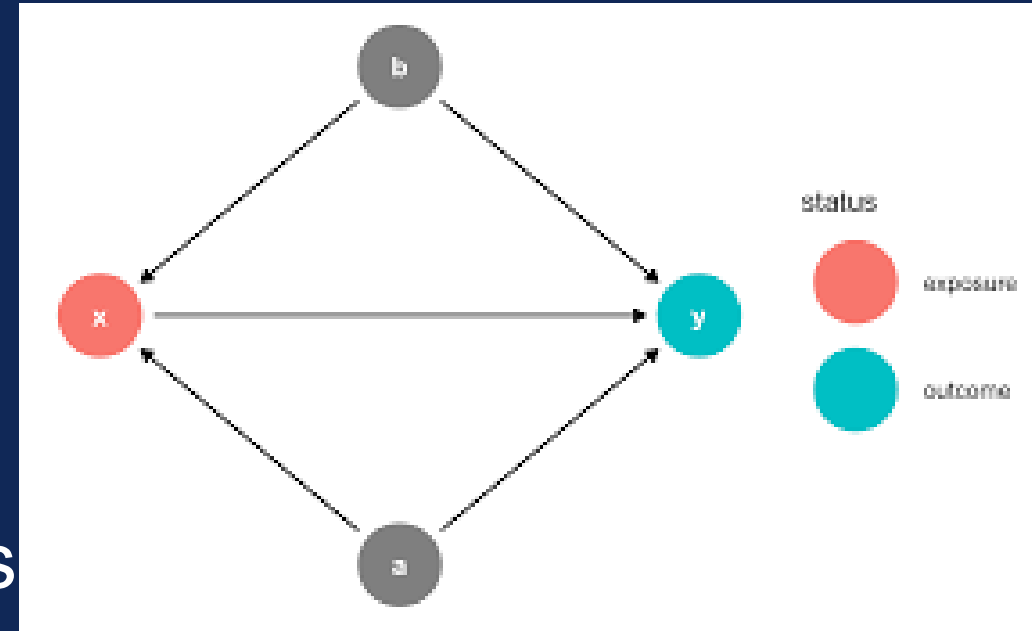


Create a structural diagram

- Relationship is detailed by a Diagonal Acyclic Graph (DAG)
- Visually represent causal relationships
- Represent **causal assumptions**.
- Identify **confounding variables**.
- Guide **statistical adjustment** strategies (e.g., adjustment)

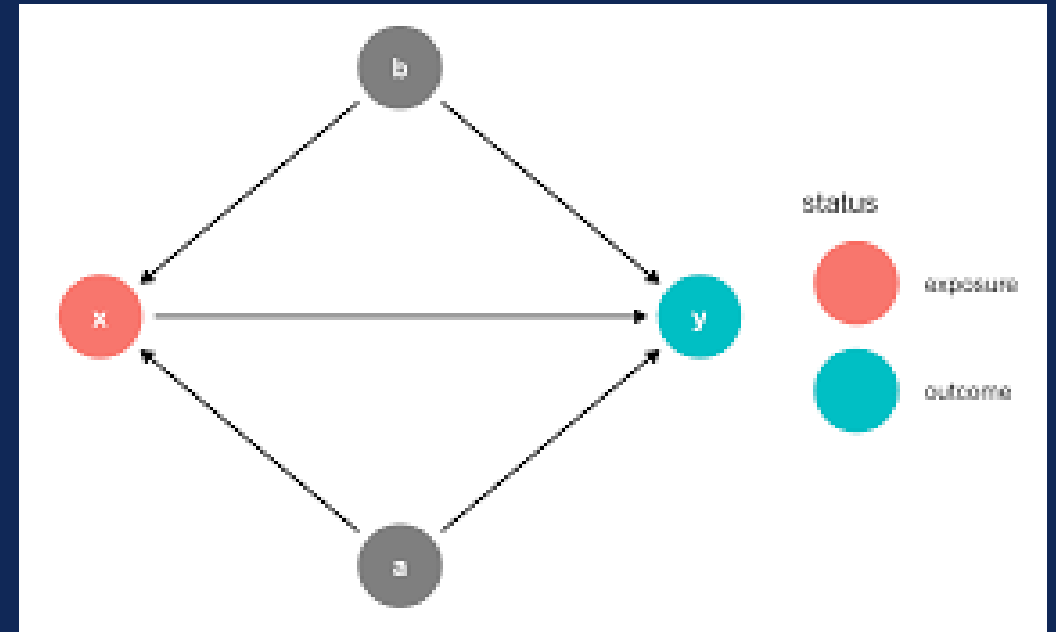
What is in a DAG?

- **Directed:** The edges (arrows) between nodes (variables) have a direction, indicating a causal or dependency relationship
• (e.g., $A \rightarrow B$ means A affects B).
- **Acyclic:** There are no loops or cycles — you cannot start at one node and follow a path that leads back to it.



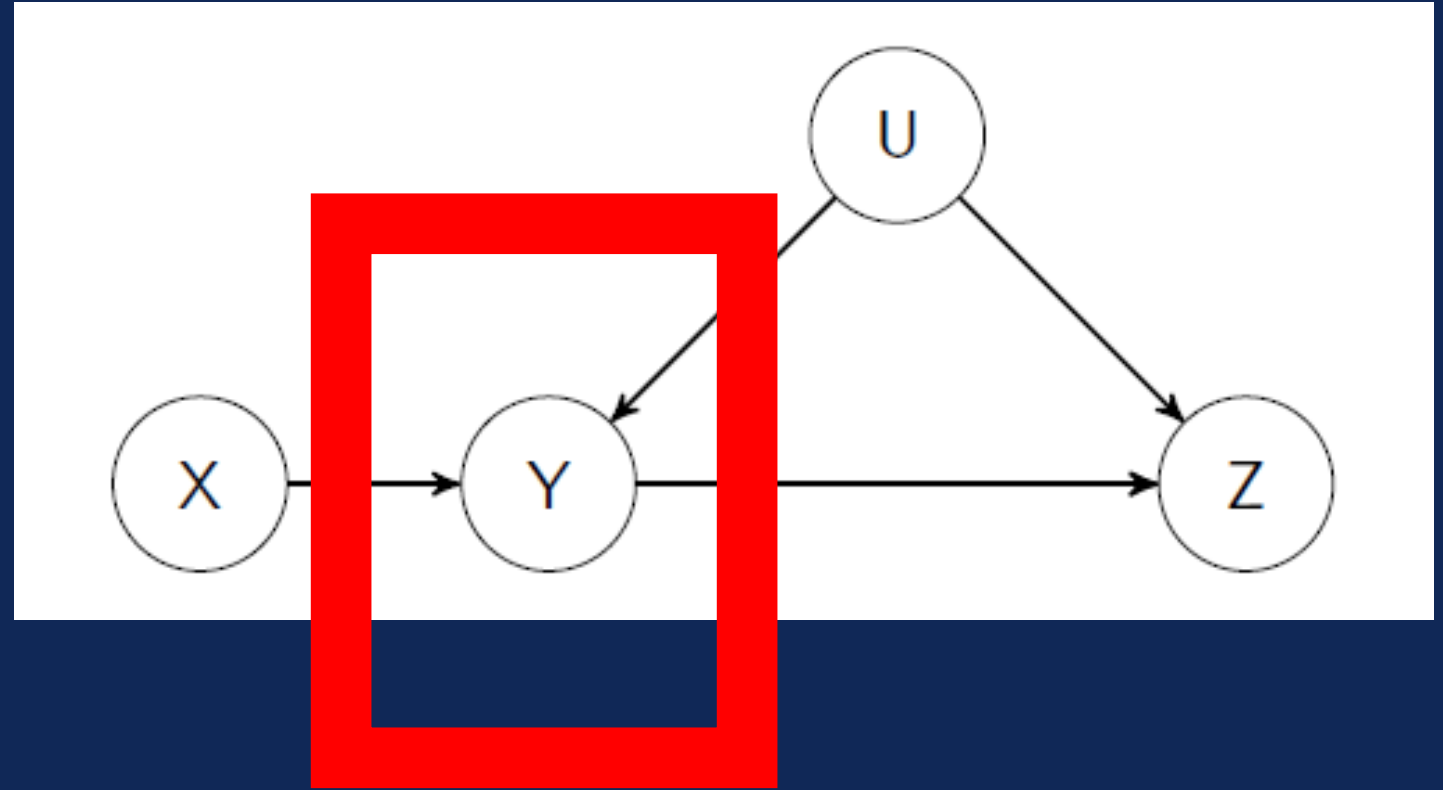
What Does a Confounder Look Like in a DAG?

- **Common Cause:** To both the exposure & outcome



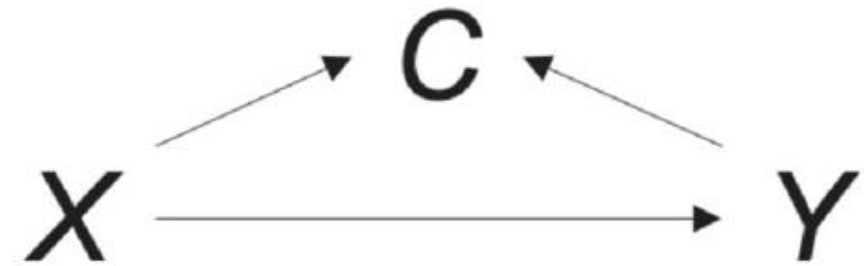
What Does a Confounder Look Like in a DAG?

- **NOT** on the pathway.



What Other Variables Can Be in a DAG?

- **Collider:** Caused by two or more other variables
- Has **incoming arrows** from two or more other variables.
- Is **not** a cause of either of those variables.
- Can introduce **bias** if conditioned on (e.g., adjusted for or stratified by).



Collider Example

The false causal relationship between athletic ability and intellectual ability

- Investigating college athletes:
 - High level college athletes are less inclined academically compared to general college students

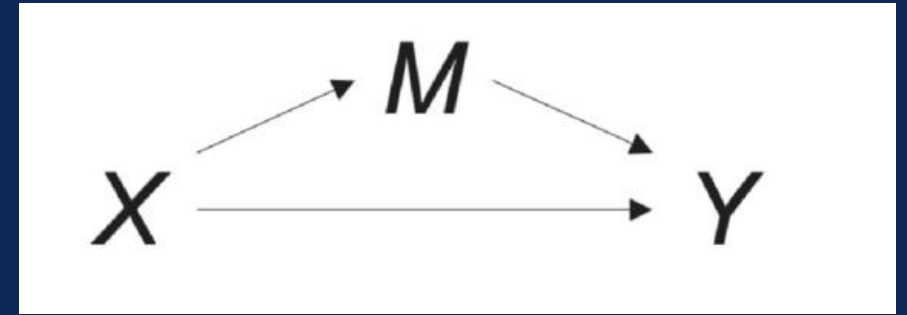
Reality

- In the general population, there is no relationship between athletic and academic ability



What Other Variables Can Be in a DAG?

- **Mediator:** A part of the causal pathway
 - Is caused by the exposure.
 - Influences the outcome.
 - Transmits part or all of the effect of the exposure to the outcome.



Mediation Example

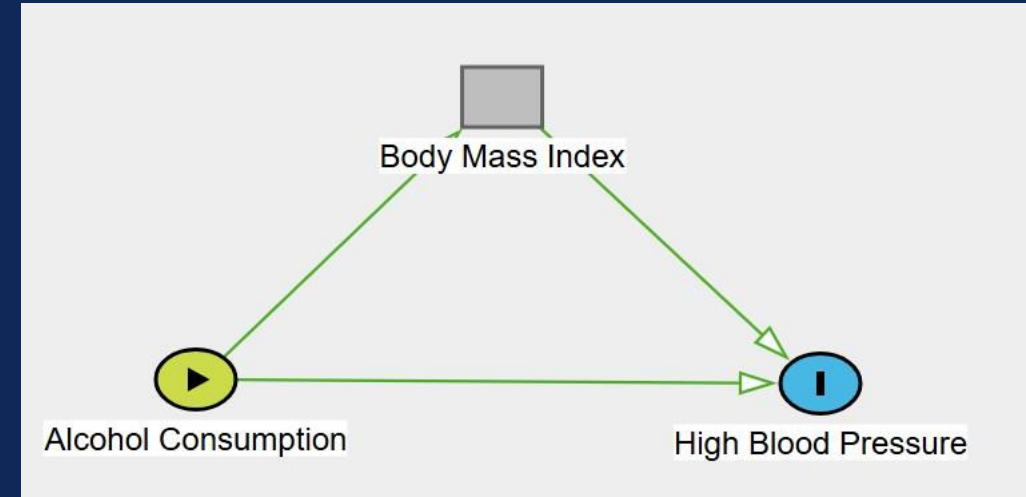
Mediator: Body Mass Index

Exposure: Alcohol Consumption

- High alcohol consumption increases body mass index (BMI)
- Also directly increases blood pressure without increase in BMI

Outcome:

High Blood Pressure



In Class Example

- Want to evaluate the causal effect of quadricep strength on ACL injury risk in female cutting and pivoting sports within the course of a season
- Exposure: Quadricep Strength
- Outcome: ACL Injury
- What are the common causes between both, that we need to account for?

Example

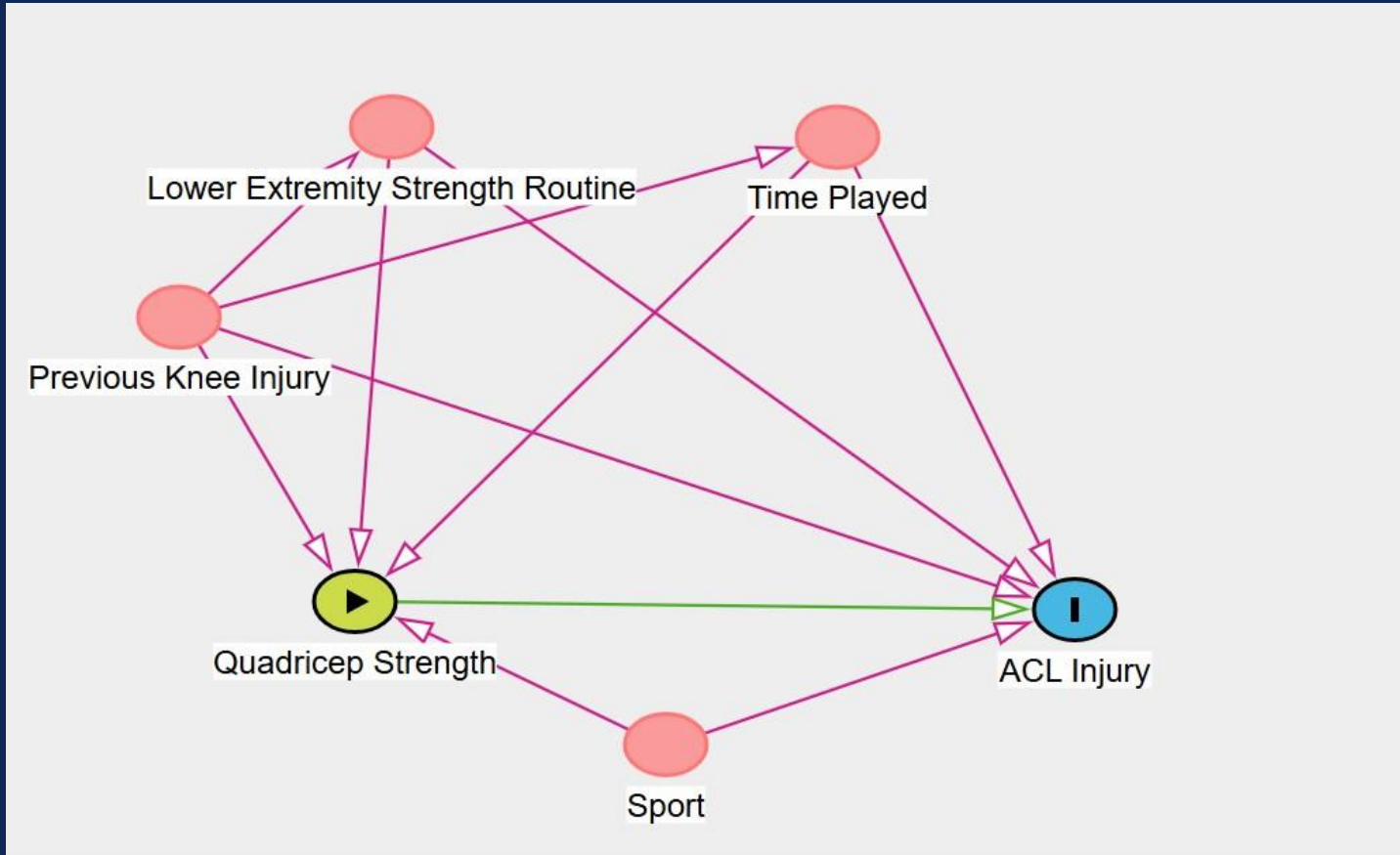
- Want to evaluate the causal effect of quadricep strengthening on ACL injury risk in female cutting and pivoting sports



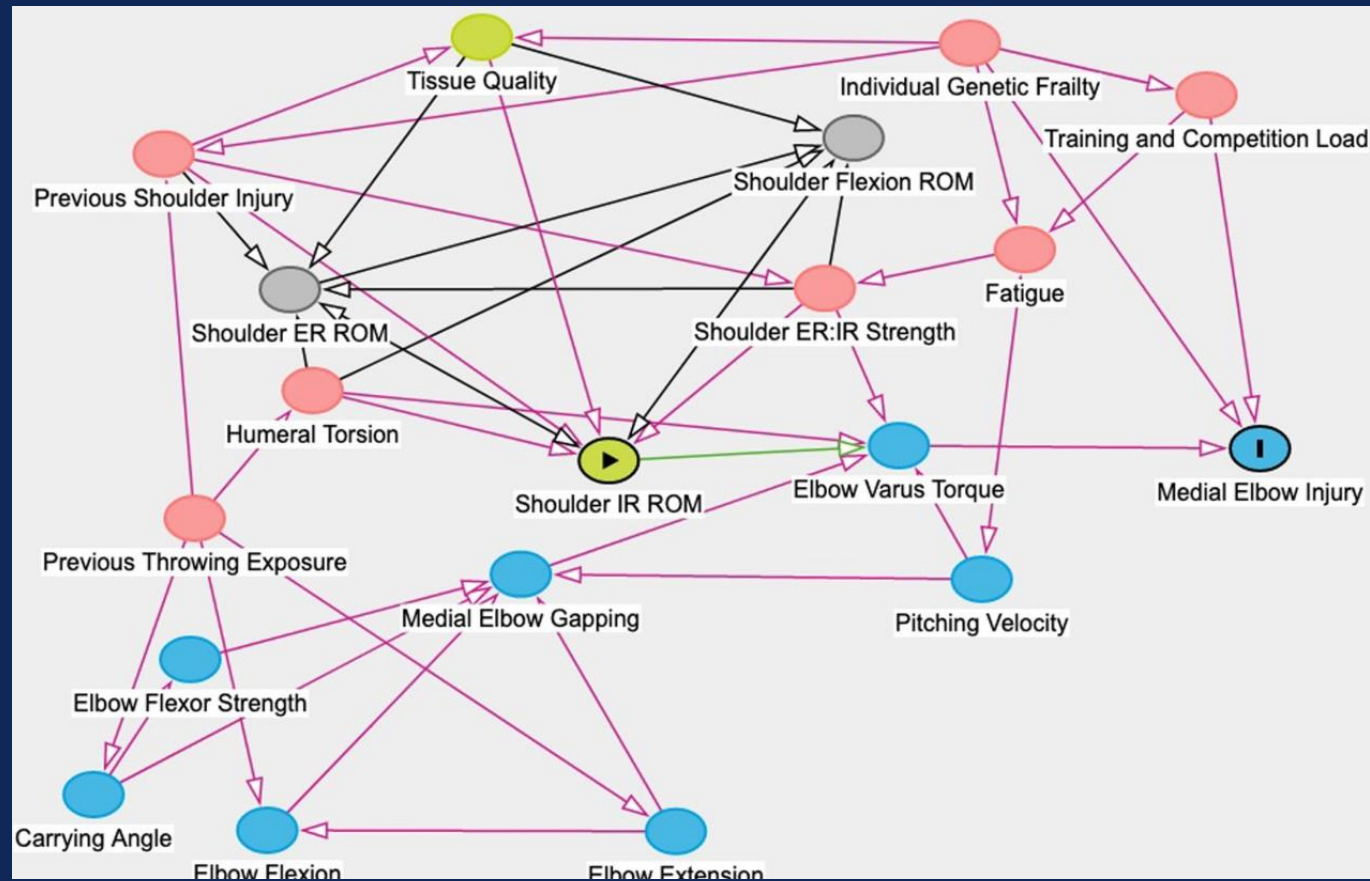
[Quadriceps Strength] → [ACL Injury Risk]

Example

- Potential Confounders: Time Played, Lower extremity strength routine, previous knee injury.



THAT WAS A SIMPLE EXAMPLE: HERE IS REAL WORLD EXAMPLE



Final Thoughts

- Controlling for confounders is essential in biomedical research
- Homes in on the signal versus the noise
- Requires a mixture of literature, expert opinion, and biological mechanisms
- Create the causal structure through DAGs
- Do not use a data driven approach.